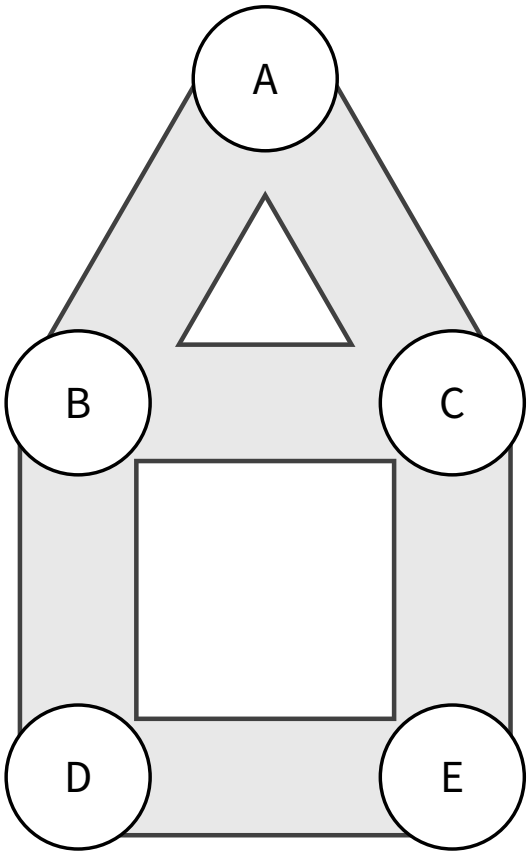
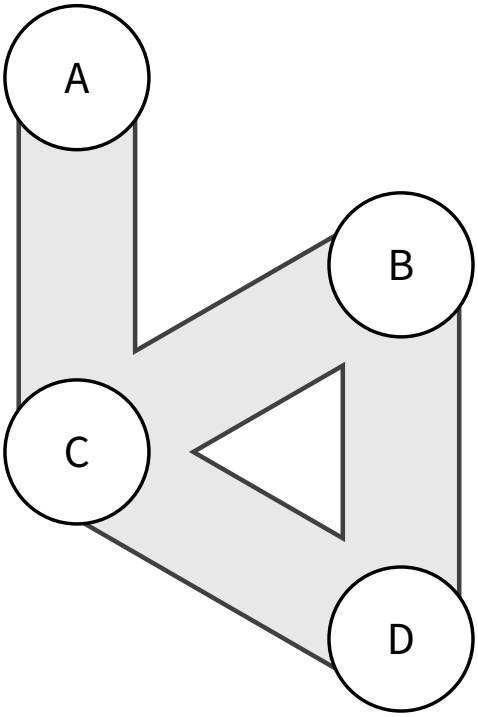
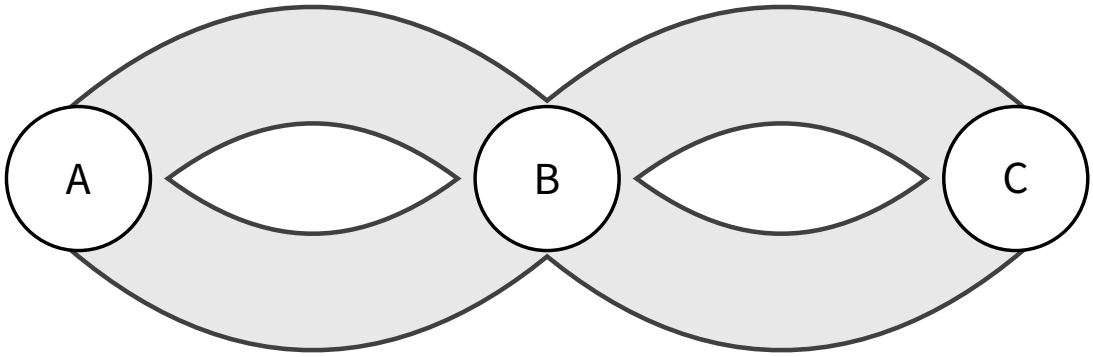
Put a counter on every bridge of a town before you walk it. A walk starts on an island and crosses bridges one after another. Each time it crosses a bridge, pick up that bridge's counter. A bridge with no counter cannot be crossed. Two islands can be joined by more than one bridge. In every town you can get from any island to any other.

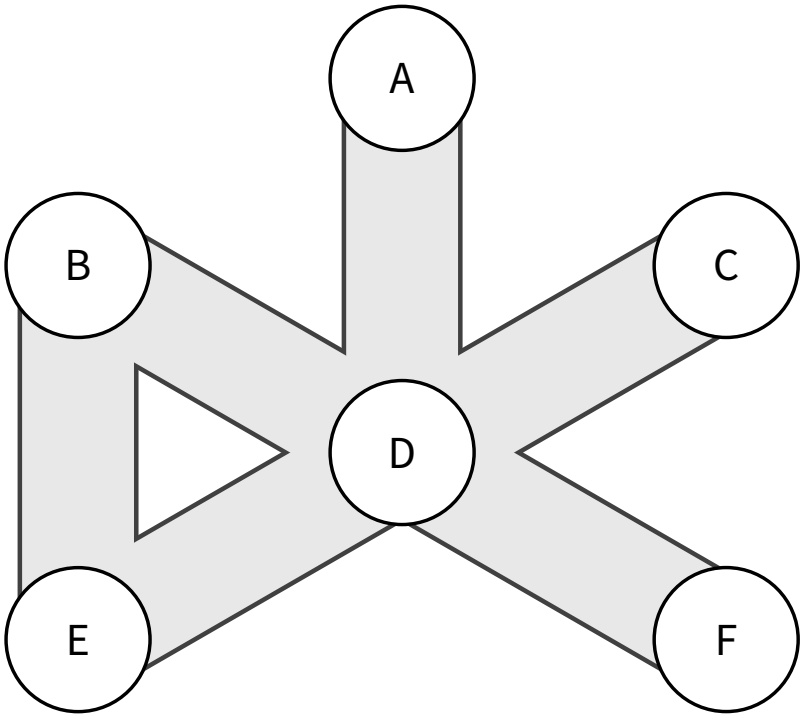
Problem 1: In each town, find a walk that picks up every counter, and write its islands in order, like A B C A.

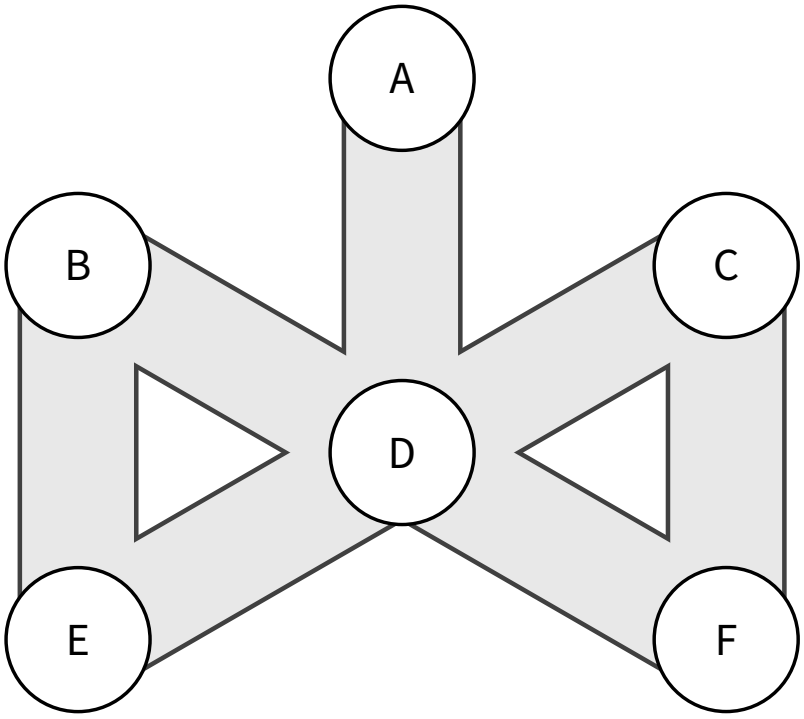


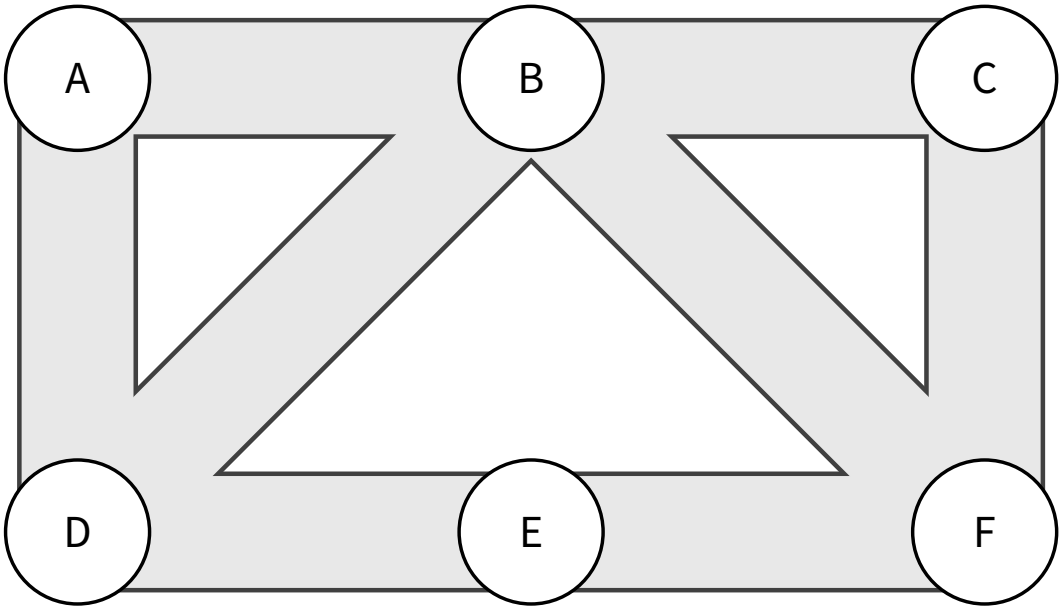


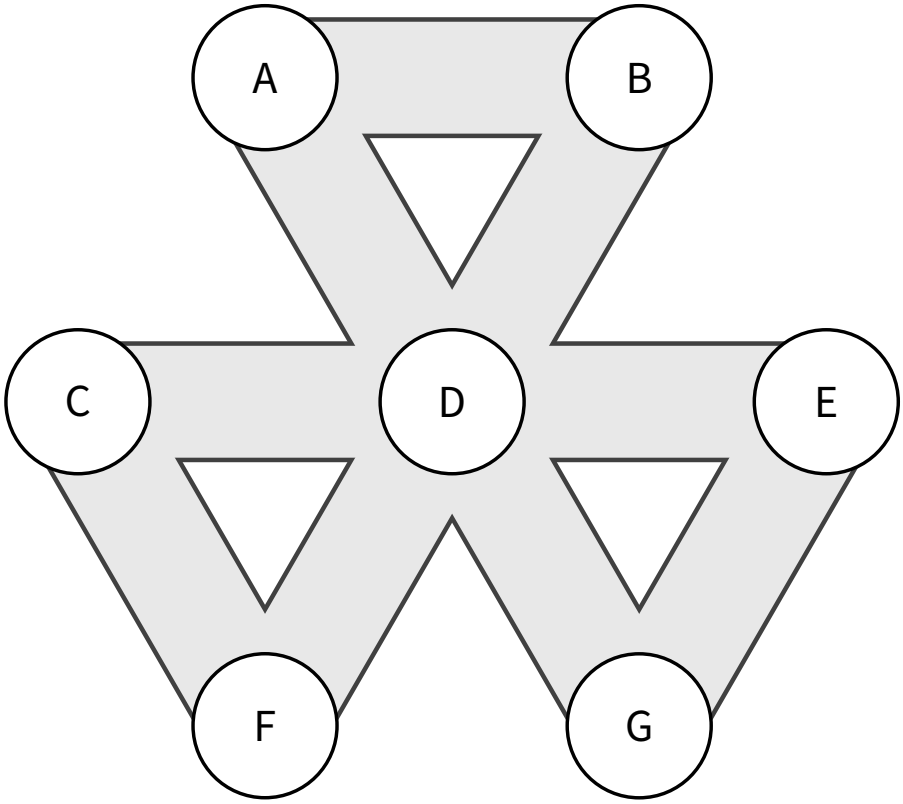


Problem 2: In each town, find every island where a walk that picks up every counter can start, and where such a walk can end. If a town has no such walk, write “none”.

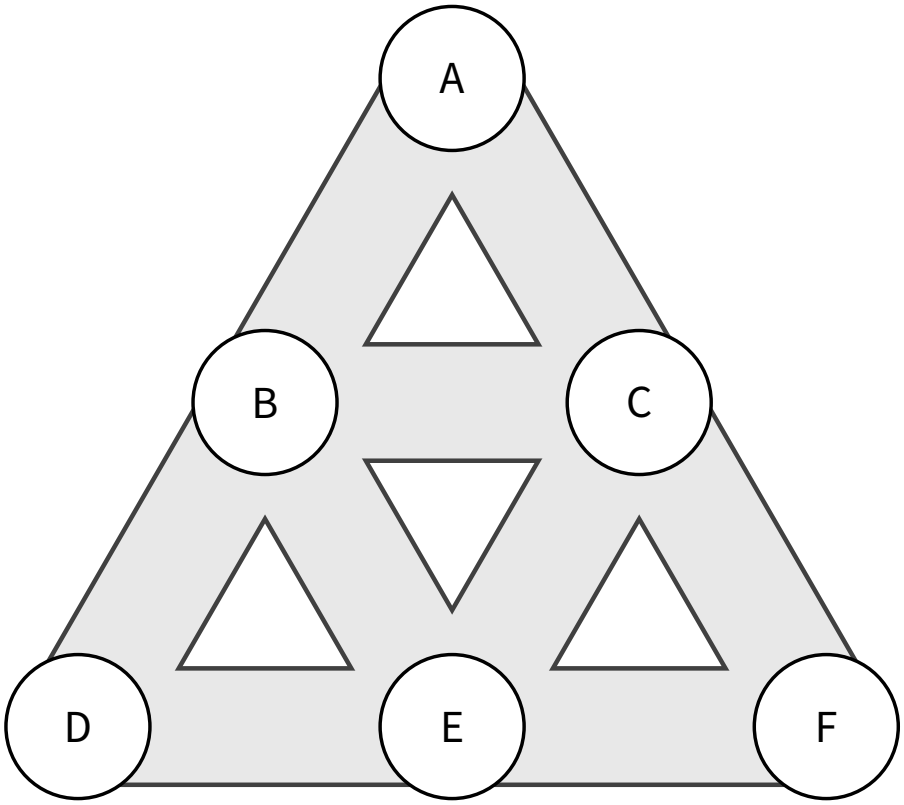


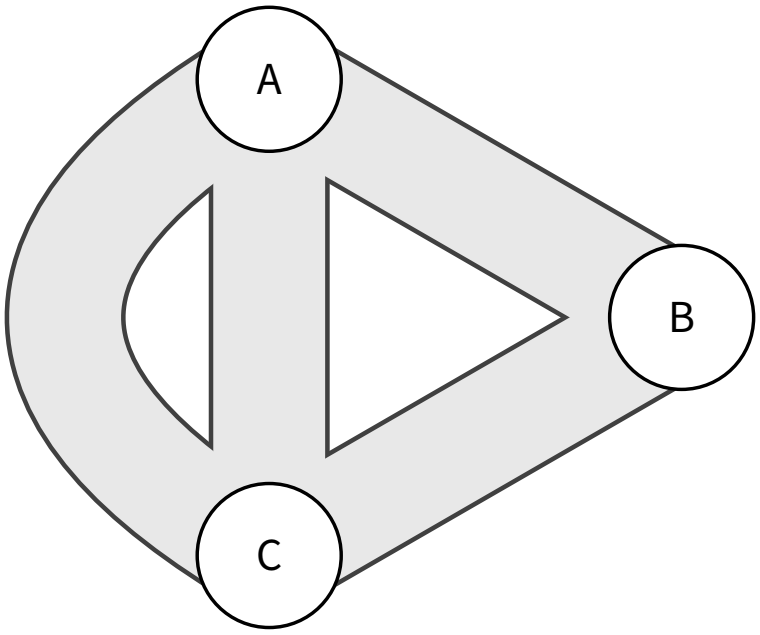


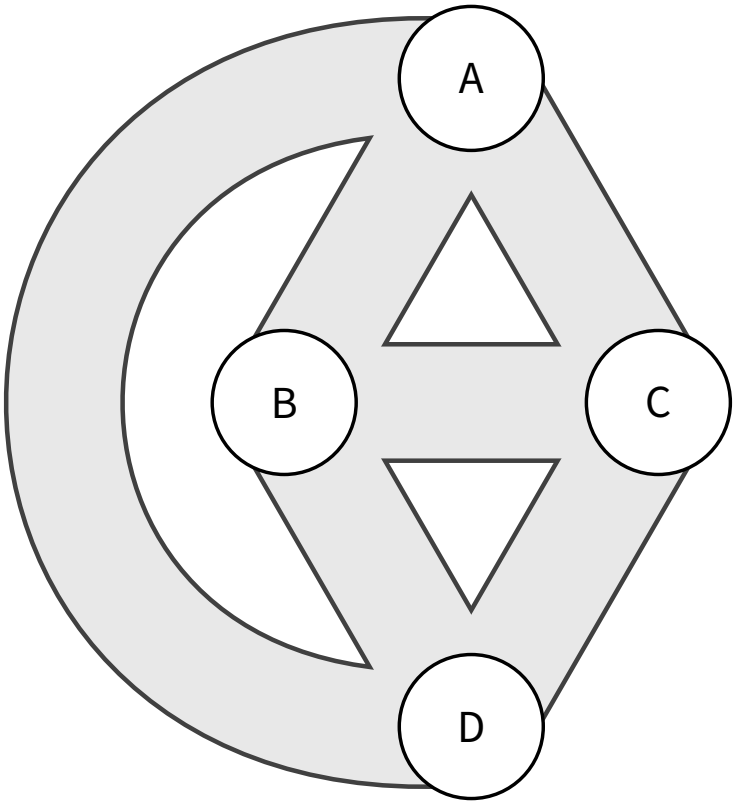

A
B
C
D
E
F

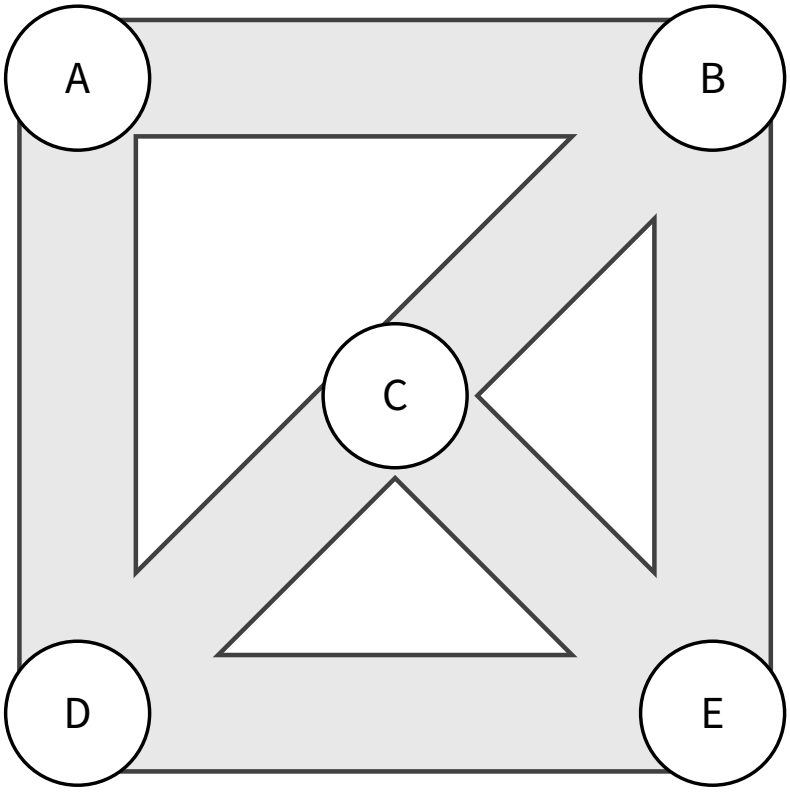

A
B
C
D
E
F
G

Problem 3: For each town, decide without walking whether a walk can pick up every counter. If it can, circle every island where such a walk can start. If it cannot, write “none”. Then check with counters.

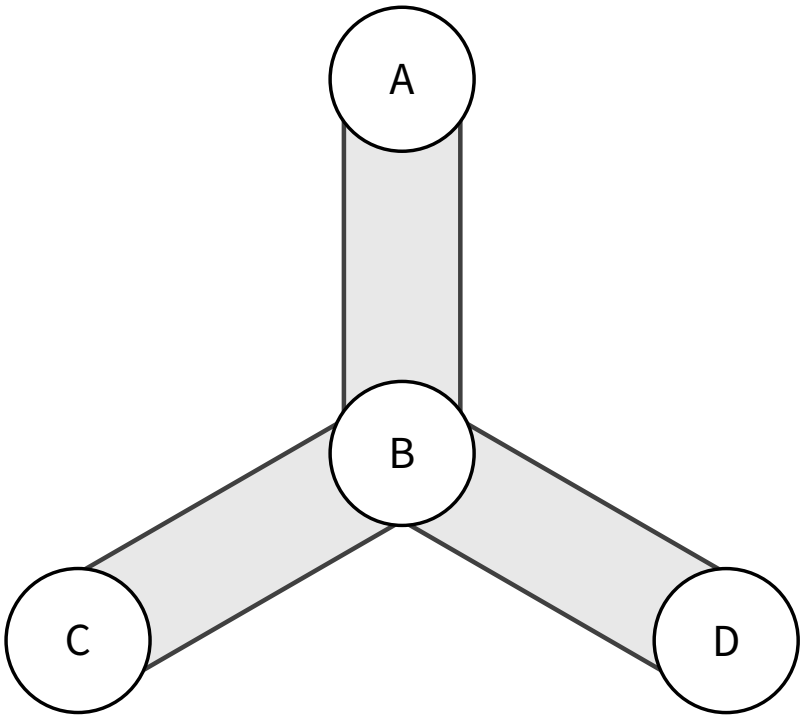


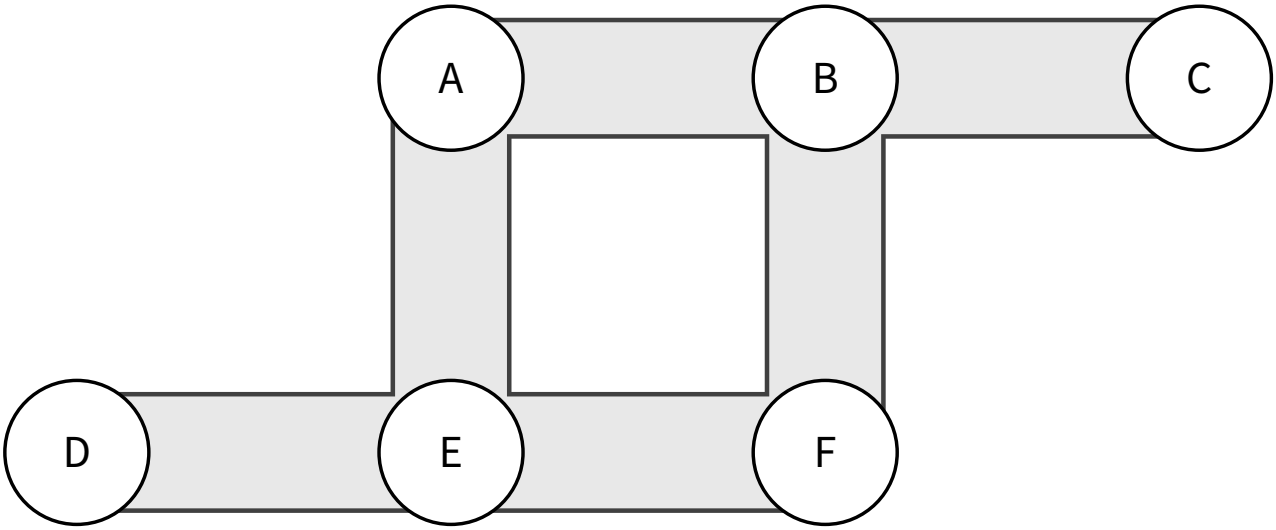


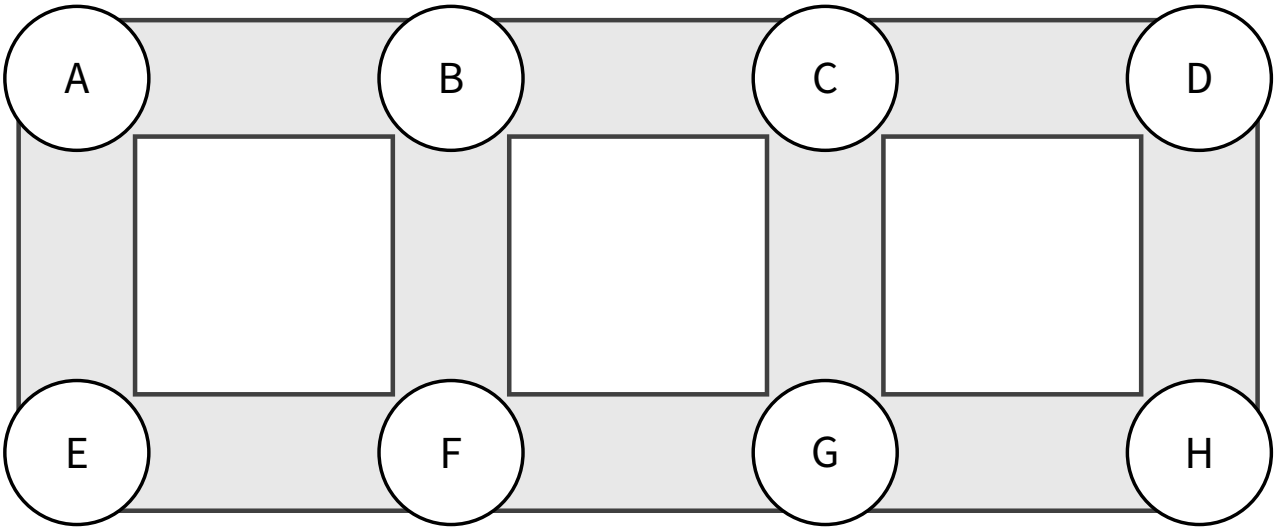

A
B
C
D


A
B
C
D
E

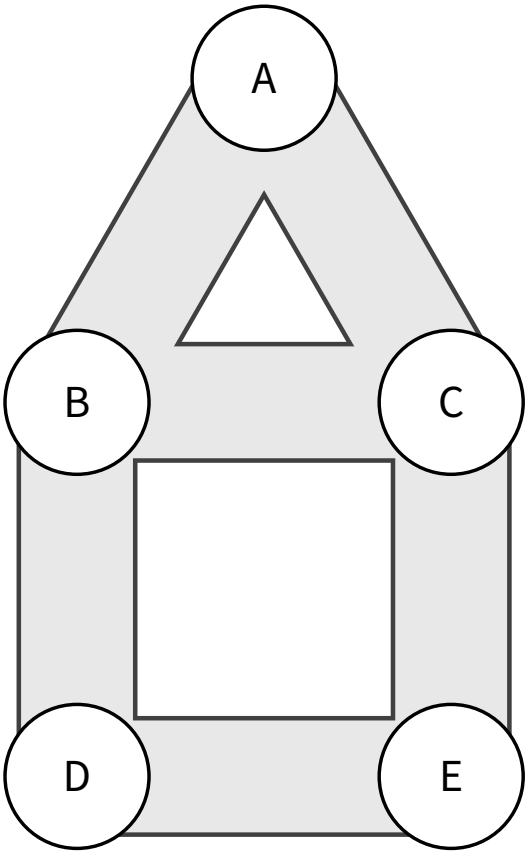
Problem 4: No walk picks up every counter in these towns. In each town, draw one new bridge so that a walk can, and write the walk.

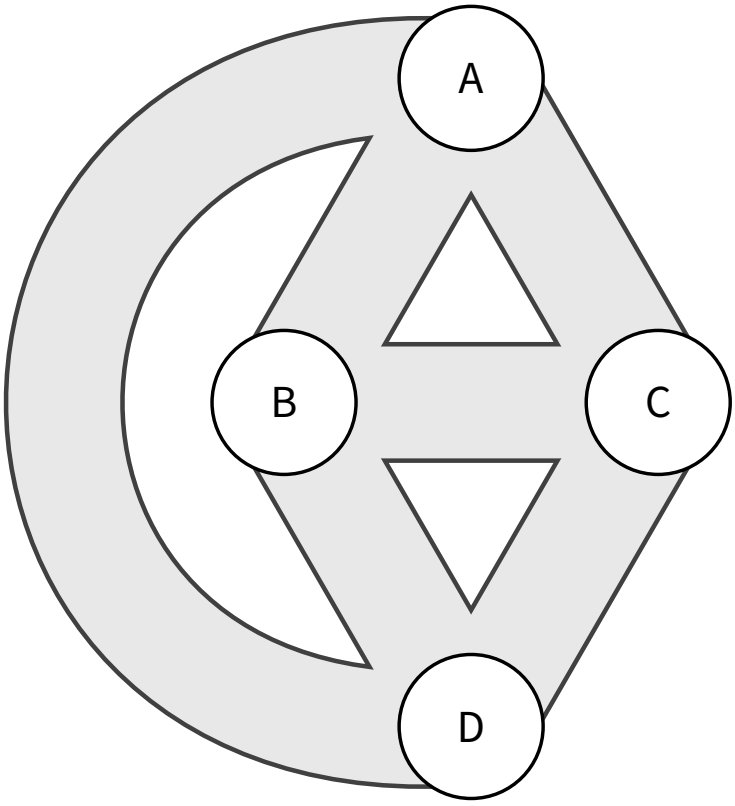




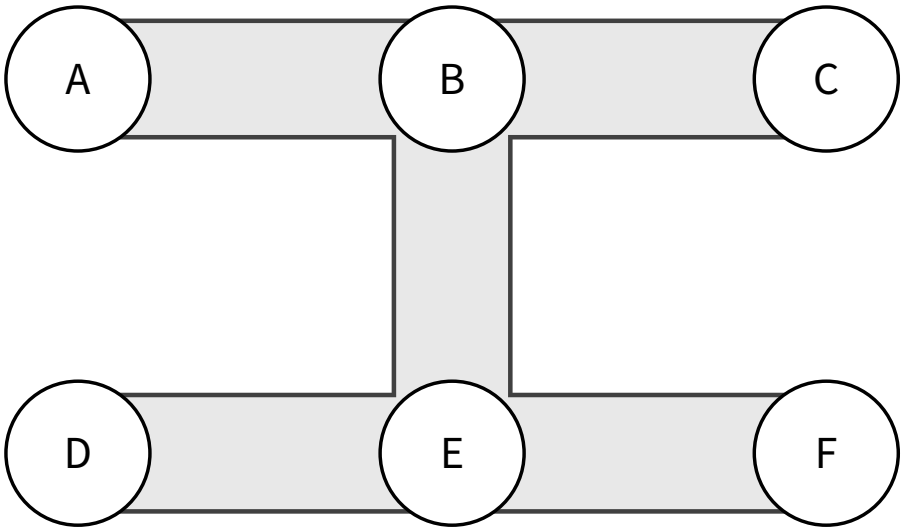


Problem 5: In each town, draw as few new bridges as you can so that a walk can pick up every counter and end on the island where it started.



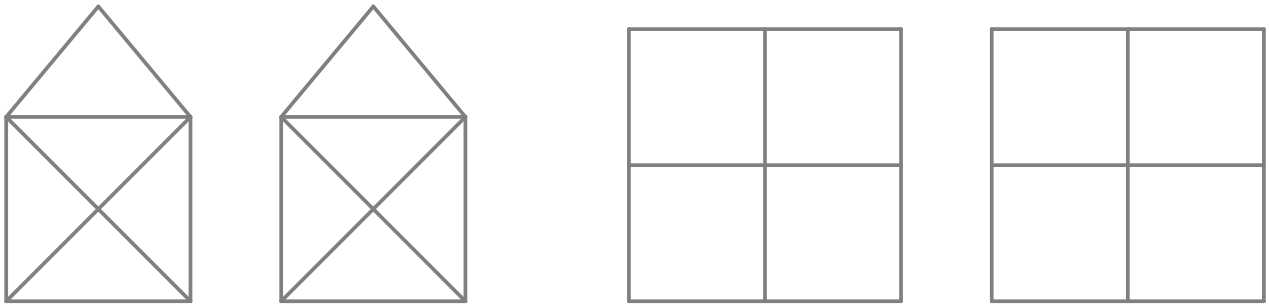


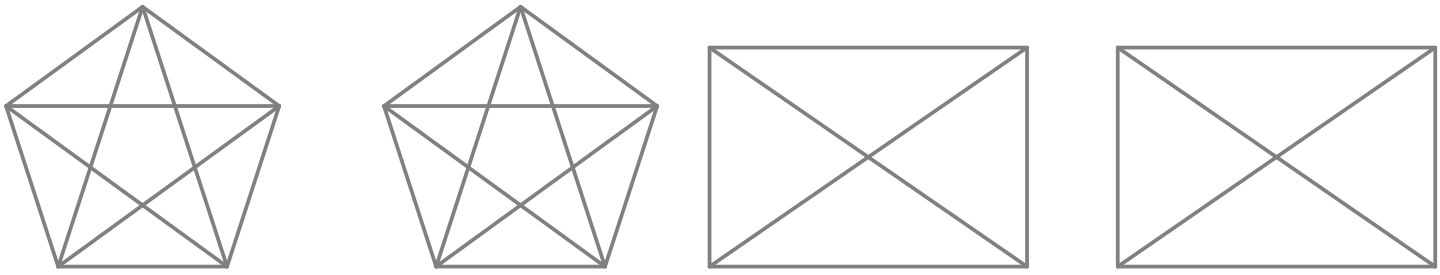
Problem 5, continued



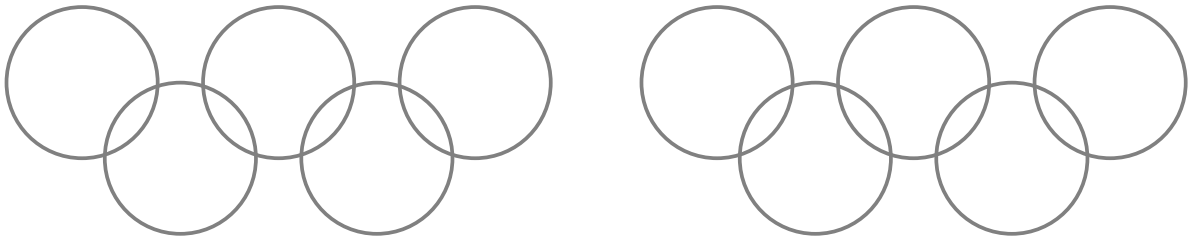
Problem 6: How can you tell, without walking, whether a town has a walk that picks up every counter, and where such a walk can start and end?

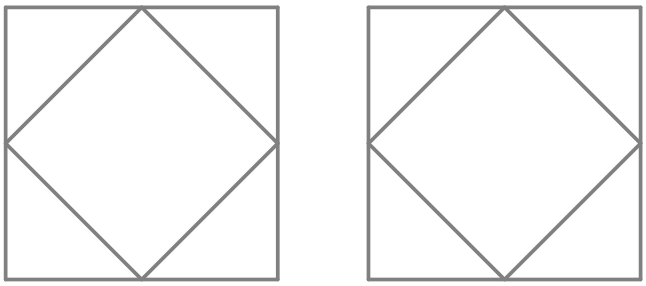
Problem 7: Trace each picture with a colored pencil, without lifting the pencil and without going over a line twice. If a picture cannot be traced this way, explain how you know.





Problem 7, continued





Problem 8: Build each of these towns with hexagons and craft sticks for your partner to walk. Draw each town you build.

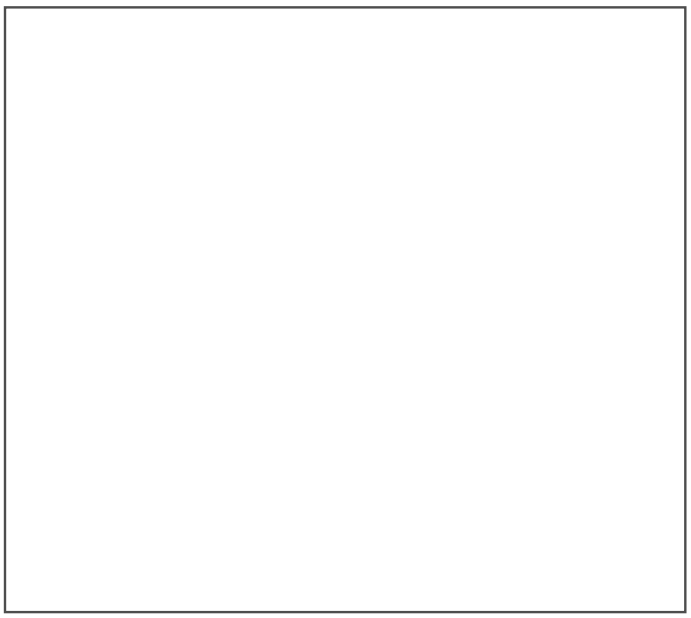
4 islands and 6 bridges. A walk that picks up every counter can start on any island.



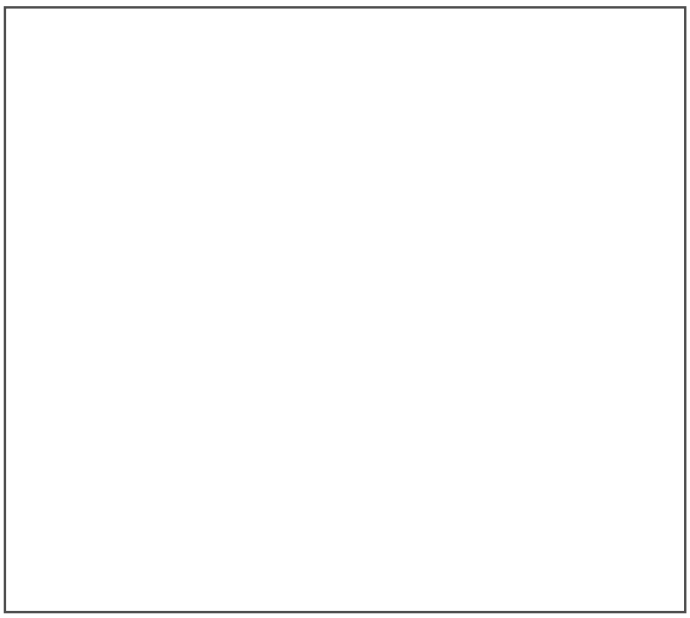
5 islands and 5 bridges. No walk picks up every counter.



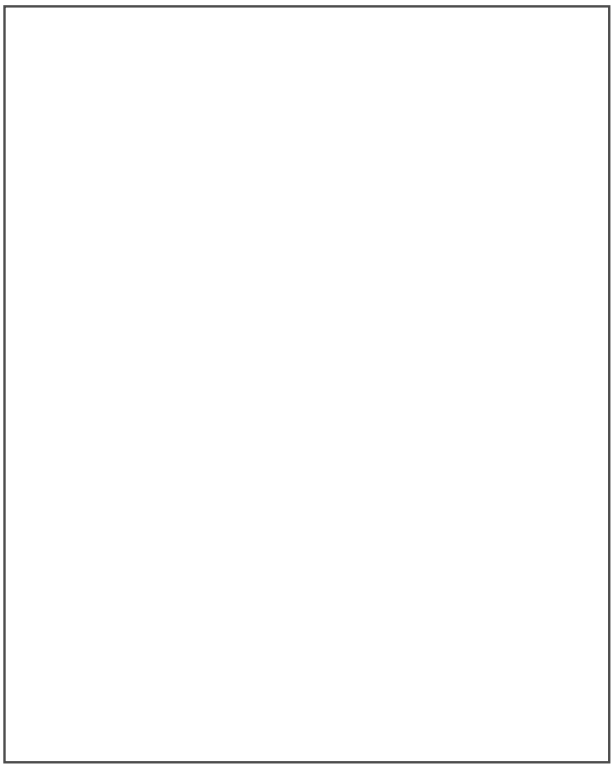
6 islands and 8 bridges. A walk that picks up every counter can start only on A or on F.



3 islands and 5 bridges. A walk that picks up every counter can start only on A or on B.



Problem 9: Can you make a town with at least one bridge where a walk that picks up every counter can start on only one island? Make one, or explain why it cannot be done.



Problem 10: A bridge can hold two counters, and then a walk can cross it two times. In each town, put a second counter on as few bridges as you can so that a walk can pick up every counter and end on the island where it started. Color each bridge that gets a second counter.

